

Atharv Khare

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Education

Indian Institute of Technology Madras

CGPA: 9.15

Bachelor of Science: Data Science & Applications

Expected 2027

Relevant Coursework: Statistical Inference, Machine Learning, Database Management Systems, Business Analytics, Data Structures & Algorithms, Probability & Statistics

Experience

Learning Assistant: Tools in Data Science, IIT Madras

Jan 2026 - May 2026

- Selected as **Learning Assistant** for IIT Madras's Tools in Data Science course; mentored 1,000+ students through technical support, debugging sessions and community initiatives.
- Built **TDS-Collab** and **TDS-Summary**, improving assignment discovery, peer collaboration, and access to course resources.

Data Analyst: Shri Melan FF (QSR), Bhopal

Sep 2025 - Feb 2026

- Designed a **market basket analytics engine** using statistical heuristics and domain-specific recommendation rules across a 153-item menu, increasing average ticket size by **14%**.
- Developed **SARIMAX + Bayesian demand forecasting** and a **BCG-based Menu Matrix**, reducing stockouts by **35%** and improving gross margin by **8.5%**.

Projects

Sentio: RAG-Powered LLM Reflection Platform 🐙GitHub 🚀Live

2026

- Built a **FastAPI + PostgreSQL/pgvector** LLM application combining Claude, semantic retrieval, Cohere reranking, and longitudinal memory for context-aware responses.
- Designed a **RAG pipeline** using Sentence-Transformers embeddings, cosine retrieval, cross-encoder reranking, and SSE token streaming with graceful fallbacks.
- Implemented **three-tier memory, structured LLM workflows, background NLP processing, and seven deterministic Socratic algorithms** for adaptive conversations.

Context Shift LogLinear Network for Label Noise Robust Classification 🐙GitHub 🚀Live

2026

- Proposed **CS-LLN**, a closed-form ΔR statistic detecting feature-pair correlation shifts across classes, fixing Naive Bayes's independence assumption via enriched L2 logistic regression.
- Benchmarked on **3 UCI datasets**: **63% fewer misclassifications** and $\sim 10\times$ **lower log-loss** than Gaussian NB, using **6.2x fewer engineered features** – rigorous statistical validation over raw model output.
- Validated on **213 real student academic records** (LNCT Bhopal) for at-risk prediction: **+6.13pp accuracy, 2.4x lower log-loss** vs. NB (5-fold CV); findings documented in a research preprint.

Comment Moderation Classifier: Data Quality & Drift Monitoring System 🐙GitHub 🚀Live

2026

- Ranked **Top 0.8% (21/2,744)** in a 4-class, severely imbalanced Kaggle competition (**0.835 macro-F1**) using a **7-model stacked ensemble** (LightGBM, NB-SVM, calibrated LinearSVC).
- Engineered **53 statistically validated features** across three TF-IDF spaces and applied per-class **Nelder-Mead threshold tuning** to recover a 2.8% minority class – systematic anomaly/gap detection in noisy, obfuscated text data.
- Deployed a **FastAPI QA dashboard** with a **SQLite audit trail** and **PSI-based drift monitoring** for continuous data-quality checks, plus RAG-grounded policy citations for explainable, documented decisions.

Achievements

- Awards:** Best Capstone Project – Business Data Management, IIT Madras BS Degree Program.
- Research:** Author of **4 research preprints** spanning **AI Systems, Machine Learning, Human-Computer Interaction, and Remote Sensing**: *GitSyntropy*, *CS-LLN*, *TypeState*, and *TerraHeal*. [ResearchGate]
- Competitions:** **Top-10** in three Kaggle competitions; Winner: **Anthropic CBC Spring Challenge**, **IIT Madras Inter-House Hackathon**, and **Anukriti IIT Madras**.
- Certifications:** Oracle Certified Foundations Associate: **Agentic AI**, NPTEL Distributed Systems (**Elite**), HackerRank **Advanced SQL**, DataCamp **Data Science Associate**.

Skills

Languages: Python, SQL, JavaScript

Data Analysis & QA: Anomaly Detection, Data Validation, Root-Cause Analysis, PSI Drift Monitoring, Automated QA Pipelines, Report Generation

Machine Learning: Classical ML, Ensemble Learning, Time-Series Forecasting, Statistical Inference

AI Systems: LLMs, RAG, LangGraph, LangChain, Vector Databases, Prompt Engineering, Multi-Agent Systems

Frameworks & Tools: FastAPI, Flask, PostgreSQL, pgvector, SQLite, Scikit-Learn, CatBoost, LightGBM, Pandas, NumPy, Git, docker, dbt